

Linear-Bayes

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Load library

```
library(coda)  
library(rjags)
```

```
## Linked to JAGS 4.3.1
```

```
## Loaded modules: basemod,bugs
```

```
library(knitr)  
library(purrr)  
library(tidyr)  
library(ggplot2)
```

set seed

```
set.seed(123)
```

Load Data

```
df <- read.csv('./ParisHousing.csv')  
head(df)
```

##	squareMeters	numberOfRooms	hasYard	hasPool	floors	cityCode	cityPartRange
## 1	75523	3	0	1	63	9373	3
## 2	80771	39	1	1	98	39381	8
## 3	55712	58	0	1	19	34457	6
## 4	32316	47	0	0	6	27939	10
## 5	70429	19	1	1	90	38045	3
## 6	39223	36	0	1	17	39489	8

##	numPrevOwners	made	isNewBuilt	hasStormProtector	basement	attic	garage
## 1	8	2005	0		1	4313	9005
## 2	6	2015	1		0	3653	2436
## 3	8	2021	0		0	2937	8852
## 4	4	2012	0		1	659	7141
## 5	7	1990	1		0	8435	2429
## 6	6	2012	0		1	2009	4552

##	hasStorageRoom	hasGuestRoom	price
## 1	0	7	7559082
## 2	1	2	8085990
## 3	1	9	5574642
## 4	0	3	3232561
## 5	1	4	7055052
## 6	0	1	3926647

Model 1

Set Dataframe as matrix and split dependent and independent variable

```
price <- as.matrix(df$price)
Y <- price

X <- df[, c("squareMeters", "numberOfRooms", "hasYard", "hasPool", "floors", "cityCode", "numPrevOwners", "made", "isNewBuilt", "hasStormProtector", "basement", "attic", "garage", "hasStorageRoom", "hasGuestRoom", "cityPartRange")]

names <- c("price", "Square Meters", "Number of Rooms", "Has Yard", "Has Pool", "Number of Floors", "City Code", "Number of Previous Owners", "Year Made", "is Newly Built", "Has Storm Protector", "Basement Area", "Attic Area", "Garage Area", "Has Storage Room", "Has Guest Room", "City Part Range")
```

Delete Missing Value

```
junk <- is.na(rowSums(X))
Y <- Y[!junk]
X <- X[!junk,]
```

Standardize Covariates

```
X <- as.matrix(scale(X))
```

JAGS

Put Data in JAGS Format

```
n <- length(Y)
p <- ncol(X)

data <- list(Y=Y,X=X,n=n,p=p)
params <- c("alpha","beta")
burn <- 500
n.iter <- 2000
n.chains <- 3
thin <- 5
```

Make Jags Model

```
model_string <- textConnection("model{
  # Likelihood
  for(i in 1:n){
    Y[i] ~ dnorm(alpha+mu[i],taue)
    mu[i] <- inprod(X[i,],beta[])
  }

  # Priors
  for(j in 1:p){
    beta[j] ~ dnorm(0,0.001)
  }

  alpha ~ dnorm(0,0.001)
  taue ~ dgamma(0.1, 0.1)

  # WAIC calculations
  for(i in 1:n){
    like[i] <- dnorm(Y[i],mu[i],taue)
  }

}")
```

Set Initial Value

```
# Set initial value list
inits = list()

# Set Initial value of random value
inits$alpha = rnorm(1)
for(i in 1:p) {
  inits$beta[i] = rnorm(1)
}
inits$taue = 10
```

Compile Model

```
model1 <- jags.model(model_string, data = data, n.chains=n.chains, quiet=TRUE, inits = inits)
```

Update Model

```
update(model1, burn)
```

Get Posterior Samples from the model

```
samples1 <- coda.samples(model1, variable.names=params, n.iter=n.iter, thin=thin)
```